

Compositional Generalization from First Principles

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Introduction

Compositional Generalization

- Ability to utilize a finite set of known components to understand and generate a vast array of novel combinations.
- **“infinite use of finite means”**
- Disentanglement and identifiable representation learning aim to learn succinct representations that both factorize the data space efficiently and are robust towards distributional changes
- This framework is independent of the exact distributions used for training and testing, and our assumptions on the compositional nature of the data allow us to prove generalization in a much broader setting.



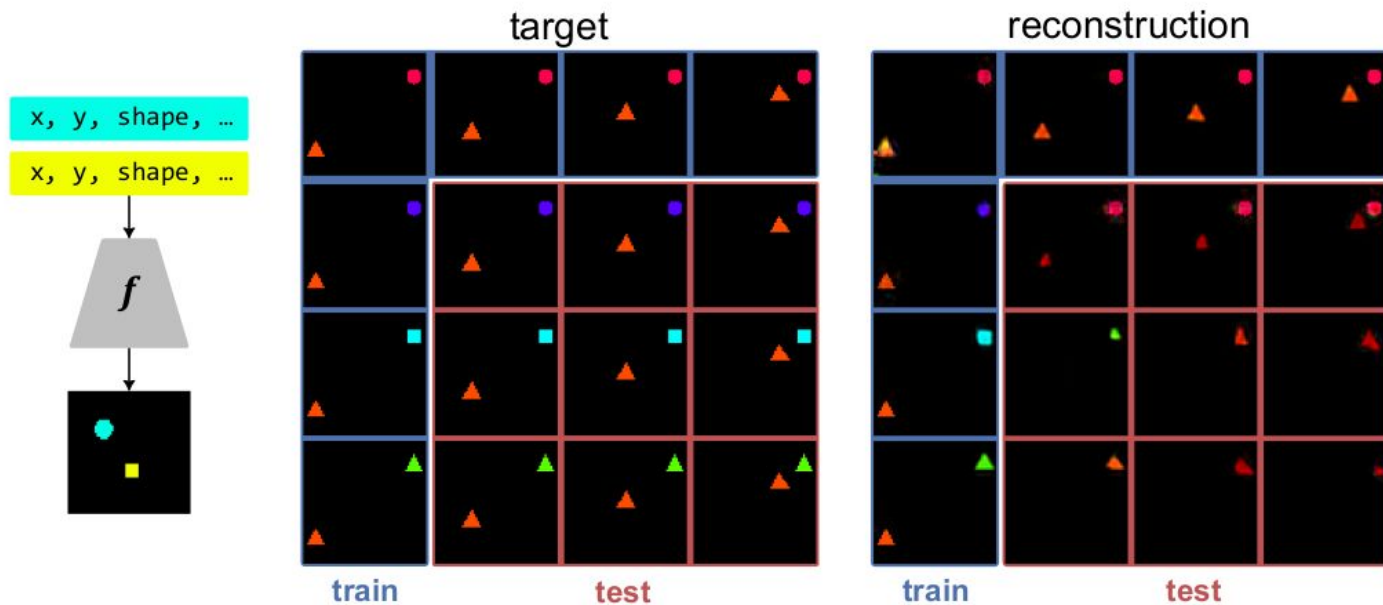
A framework for compositional generalization

Compositional Data

- Colloquially, the term “compositional data” implies that the data can be broken down into discrete, identifiable components that collectively form the whole.
- **Component** function and **composition** function.
- simple dataset similar to multi-dSprites
- Each sample in this **dataset** is a composition of **two basic sprites**, each with a random position, shape, size, and color, size.
- Samples are generated by a simple rendering **engine**.
- Composition function: a **simple pixel-wise function**

$$\mathbf{z}_1 = (z_{1,x}, z_{1,y}, z_{1,shape}, z_{1,size}, z_{1,color})$$

Compositional Data



Compositional Data

- **Attention!!!!**
- Given the latent factors, a simple multi-layer neural network can **easily learn to reconstruct** images containing compositions of these sprites that were covered by the **training set**. However, reconstruction **fails for novel compositions**, even if the individual components have been observed before.

Compositional Representation

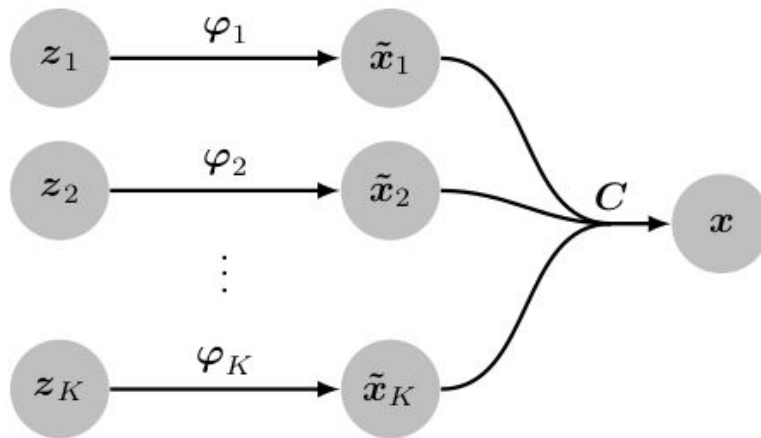
Definition 1 (Compositional representation). $\{C, \varphi_1, \dots, \varphi_K, \mathcal{Z}_1, \dots, \mathcal{Z}_K, \tilde{\mathcal{X}}_1, \dots, \tilde{\mathcal{X}}_K\}$ is a *compositional representation* of function f if

$$\forall z \in \mathcal{Z} \quad f(z) = C(\varphi_1(z_1), \dots, \varphi_K(z_K)) \quad \text{and} \quad \mathcal{Z} = \mathcal{Z}_1 \times \dots \times \mathcal{Z}_K, \quad (1)$$

where z_i denotes the canonical projection of z onto $\mathcal{Z}_i$. We refer to $\varphi_k : \mathcal{Z}_k \rightarrow \tilde{\mathcal{X}}_k$ as the *component functions*, to $\tilde{\mathcal{X}}_1, \dots, \tilde{\mathcal{X}}_K$ as the (hidden) component spaces, and to $C : \tilde{\mathcal{X}}_1 \times \dots \times \tilde{\mathcal{X}}_K \rightarrow \mathcal{X}$ as the *composition function*.

- The compositional representation of a function f is not unique. $\{f, \text{Id}, \dots, \text{Id}\}$

Compositional Representation



Compositional Support

$$\text{supp } P = \{ (z_1 \in \mathcal{Z}_1, z_2 \in \mathcal{Z}_2) | z_1 = z_1^0 \vee z_2 = z_2^0 \} .$$

Definition 2 (Compositional support). Given two arbitrary distribution P, Q over latents $z = (z_1, \dots, z_K) \in \mathcal{Z} = \mathcal{Z}_1 \times \dots \times \mathcal{Z}_K$, P has *compositional support* w.r.t. Q if

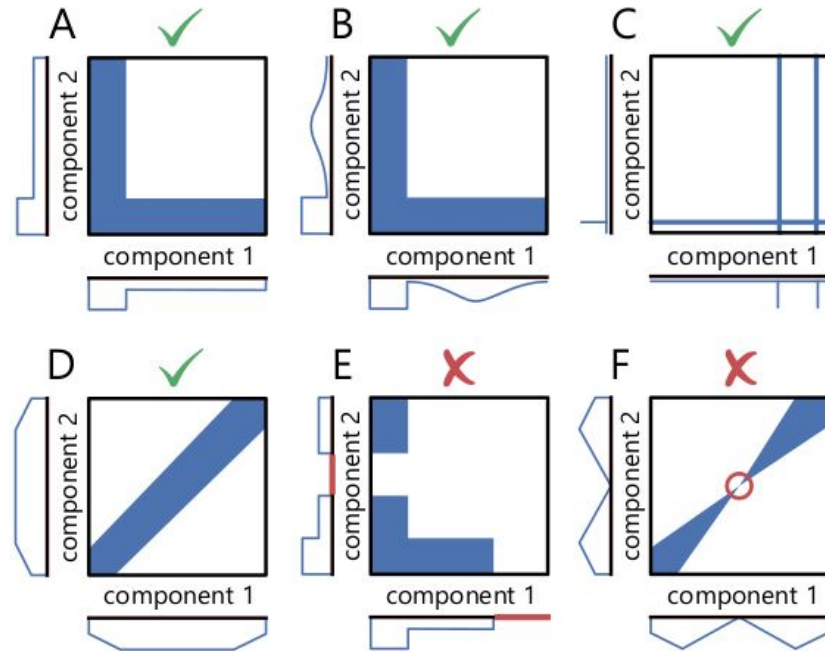
$$\text{supp } P_{z_k} = \text{supp } Q_{z_k} \subseteq \mathcal{Z}_k \quad \forall k \in [K]. \quad (3)$$



Interpolation or Extrapolation



Compositional Support



Conditions for Compositional Generalization

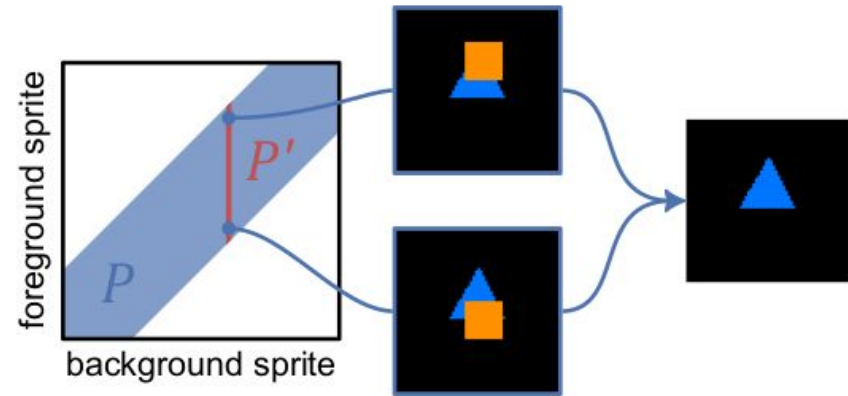
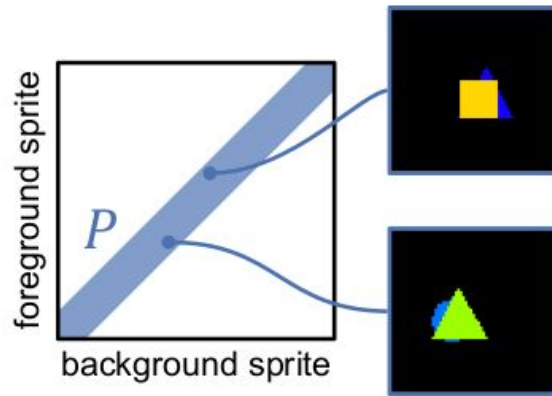
- We can reconstruct Jacobian Matrix.
- If $\text{supp } P$ is an openset.
 - A subset U of a metric space (M, d) is called *open* if, for any point x in U , there exists a real number $\varepsilon > 0$ such that any point $y \in M$ satisfying $d(x, y) < \varepsilon$ belongs to U .
Equivalently, U is open if every point in U has a neighborhood contained in U .
- With a linear composition function.
- What about constant of integration?

$$f(z) = \varphi_1(z_1) + \varphi_2(z_2).$$

$$\frac{\partial f}{\partial z_k}(z) = \frac{\partial \varphi_k}{\partial z_k}(z_k).$$

$$\hat{f} \equiv_Q f?$$

Sufficient support



Sufficient support

Definition 3 (Sufficient support). A distribution P over latents $\mathbf{z} = (z_1, \dots, z_K) \in \mathcal{Z}$, has *sufficient support* w.r.t. a compositional representation of a function $\mathbf{f}$, if $\text{supp } P$ is in an open set and for any latent value $\mathbf{z}_k^*$, there exists a (finite) set of points $P'_k(\mathbf{z}_k^*) \subseteq \{\mathbf{p} \in \text{supp } P | \mathbf{p}_k = \mathbf{z}_k^*\}$ for which the sum of total derivatives of $\mathbf{C}$ has full rank. That is,

$$\text{rank} \sum_{\mathbf{p} \in P'_k(\mathbf{z}_k^*)} \frac{\partial \mathbf{C}}{\partial \varphi_k}(\varphi(\mathbf{p})) = M, \quad (6)$$

where M is the dimension of the component space $\tilde{\mathcal{X}}_k \subseteq \mathbb{R}^M$.

Theorem

Theorem 1. *Let P, Q be arbitrary distributions over latents $\mathbf{z} = (z_1, \dots, z_K) \in \mathcal{Z}$. Let $\mathbf{f}, \hat{\mathbf{f}}$ be functions with compositional representations in the sense of definition 1 that share $\{\mathbf{C}, \mathcal{Z}_1, \dots, \mathcal{Z}_K\}$, but use arbitrary $\{\varphi_1, \dots, \varphi_K, \tilde{\mathcal{X}}_1, \dots, \tilde{\mathcal{X}}_K\}, \{\hat{\varphi}_1, \dots, \hat{\varphi}_K, \hat{\mathcal{X}}_1, \dots, \hat{\mathcal{X}}_K\}$.*

Assume the following assumptions hold:

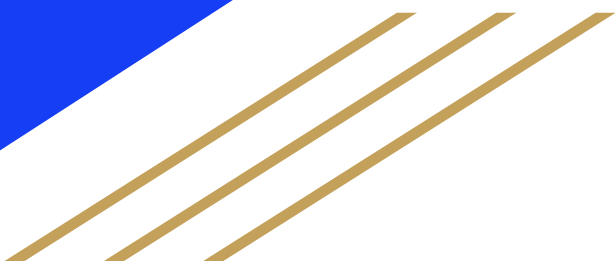
(A1) $\mathbf{C}, \varphi_k, \hat{\varphi}_k$ are differentiable, $\mathbf{C}$ is Lipschitz in φ , and φ is continuous in $\mathbf{z}$.

(A2) P has compositional support w.r.t. Q in the sense of definition 2.

(A3) P has sufficient support w.r.t. $\mathbf{f}$ in the sense of definition 3.

(A4) There exists an initial point $\mathbf{p}^0 \in \text{supp } P$ such that $\varphi(\mathbf{p}^0) = \hat{\varphi}(\mathbf{p}^0)$.

Then $\hat{\mathbf{f}}$ generalizes to Q , i.e. $\mathbf{f} \equiv_P \hat{\mathbf{f}} \implies \mathbf{f} \equiv_Q \hat{\mathbf{f}}$.



Experiments

Setup

- Compositional Model
- Monolithic Model
 - Same scale
- Composition Function $\mathbf{x} = \sigma(\tilde{\mathbf{x}}_1) \cdot \tilde{\mathbf{x}}_1 + \sigma(-\tilde{\mathbf{x}}_1) \cdot \tilde{\mathbf{x}}_2,$
 - Allow Occlude
- Both models have the capacity to fit the data by training on random samples.
- Both models can reconstruct ID samples, but only the compositional model generalizes to the entire latent space

Results

#	Train Set	Model	MSE ID $\downarrow$	MSE all $\downarrow$	R^2 ID $\uparrow$	R^2 all $\uparrow$
1	Random	Monolithic	$1.73\text{e-}3_{\pm 1.5\text{e-}5}$	$1.73\text{e-}3_{\pm 1.5\text{e-}5}$	$0.931_{\pm 5.8\text{e-}4}$	$0.931_{\pm 5.8\text{e-}4}$
2	Random	Compositional	$1.07\text{e-}3_{\pm 2.6\text{e-}5}$	$1.07\text{e-}3_{\pm 2.6\text{e-}5}$	$0.957_{\pm 1.0\text{e-}3}$	$0.957_{\pm 1.0\text{e-}3}$
3	Orthogonal	Monolithic	$8.49\text{e-}4_{\pm 2.9\text{e-}5}$	$4.06\text{e-}2_{\pm 3.9\text{e-}3}$	$0.948_{\pm 1.7\text{e-}3}$	$-0.500_{\pm 6.7\text{e-}2}$
4	Orthogonal	Compositional	$6.94\text{e-}4_{\pm 1.1\text{e-}5}$	$1.24\text{e-}3_{\pm 4.1\text{e-}5}$	$0.957_{\pm 6.4\text{e-}4}$	$0.951_{\pm 1.4\text{e-}3}$
5	Ortho. $\sim \mathcal{N}$	Compositional	$7.01\text{e-}4_{\pm 8.7\text{e-}6}$	$1.24\text{e-}3_{\pm 2.6\text{e-}5}$	$0.957_{\pm 5.5\text{e-}4}$	$0.951_{\pm 1.0\text{e-}3}$
6	Diagonal	Compositional	$8.87\text{e-}4_{\pm 1.0\text{e-}4}$	$1.39\text{e-}3_{\pm 4.0\text{e-}4}$	$0.954_{\pm 5.4\text{e-}3}$	$0.945_{\pm 1.6\text{e-}2}$
7	Ortho. (broad)	Compositional	$6.50\text{e-}4_{\pm 1.5\text{e-}5}$	$1.16\text{e-}3_{\pm 3.3\text{e-}5}$	$0.959_{\pm 9.7\text{e-}4}$	$0.954_{\pm 1.3\text{e-}3}$
8	Diag. (broad)	Compositional	$9.51\text{e-}4_{\pm 2.8\text{e-}5}$	$1.13\text{e-}3_{\pm 3.1\text{e-}5}$	$0.957_{\pm 1.2\text{e-}3}$	$0.955_{\pm 1.2\text{e-}3}$
9	Ortho. (gap)	Compositional	$7.36\text{e-}4_{\pm 2.7\text{e-}5}$	$2.64\text{e-}3_{\pm 1.8\text{e-}4}$	$0.954_{\pm 1.7\text{e-}3}$	$0.895_{\pm 7.4\text{e-}3}$
10	Diag. (narrow)	Compositional	$2.22\text{e-}3_{\pm 1.2\text{e-}3}$	$1.04\text{e-}2_{\pm 7.1\text{e-}3}$	$0.867_{\pm 7.5\text{e-}2}$	$0.589_{\pm 2.8\text{e-}1}$
11	Orthogonal	Comp. (RGBa)	$2.67\text{e-}4_{\pm 2.8\text{e-}6}$	$5.24\text{e-}4_{\pm 3.7\text{e-}6}$	$0.984_{\pm 1.8\text{e-}4}$	$0.979_{\pm 1.4\text{e-}4}$

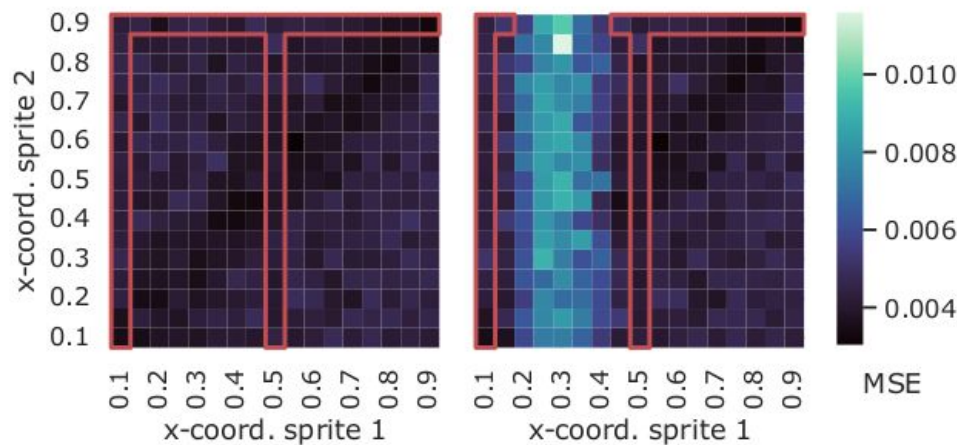
Results

- Flexible compositional support
- broadening the support of does not further increase performance.

5	Ortho. $\sim \mathcal{N}$	Compositional	$7.01\text{e-}4_{\pm 8.7\text{e-}6}$	$1.24\text{e-}3_{\pm 2.6\text{e-}5}$	$0.957_{\pm 5.5\text{e-}4}$	$0.951_{\pm 1.0\text{e-}3}$
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Violating Conditions

- Gaps in support
- Insufficient training variability
- Collapsed Composition Function



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Any Question?

Thank you for your attention!



Reference

- Thaddäus Wiedemer, Prasanna Mayilvahanan, Matthias Bethge, Wieland Brendel.
Compositional Generalization from First Principles (NeurIPS 2023)